The Ukraine conflict has seen the emergence of Unmanned Aerial vehicles (UAVs) or drones as a potent military weapon. Along with traditional drones, another lethal, unmanned weapon was unleashed in late October 2022, when, in a bold move, the Ukrainian navy attacked Russia's Black Sea Fleet headquarters in Sevastopol, on the Crimean peninsula, with seven unmanned surface vessels (USVs)—each 5.5 metre long, with a gross weight of 1,000 kg, and packed with explosives and missiles. Two more attacks were attempted on Russian naval assets in March and April. Though the Russian Navy repelled all three attacks, they announced the advent of USVs as a force in naval warfare. Not just USVs, the navies of the US, UK, China and France are rapidly adding more variety in their unmanned armoury—Naval Unmanned Aerial Systems (NUAS) and Unmanned Underwater Vehicles (UUVs). NUAS are advanced drones, but are ship-borne and integrated with the host ship's system, with launch and recovery capabilities like in a carrier with its aircraft. UUVs, also known as underwater drones, operate under the water surface and are akin to unmanned submarines. USVs, UUVs and NUAS can be either remote-controlled or autonomous. The latter involves extensive use of Artificial Intelligence (AI), crucial for functions like autonomous navigation and obstacle recognition and avoidance, and the use of sophisticated cameras and sensors. Though all three can be designed to carry a munitions payload and used to strike enemy warships, till now they have mostly been used for intelligence, surveillance and reconnaissance (ISR). Non-military uses of these unmanned craft include oceanographic and hydrographic surveys. While unmanned aerial vehicles or drones carry sensors, target designators, transmitters, GPS, autopilot systems and AI technology, USVs and UUVs have computer software/ AI tied to payload requirements, power and propulsion, hull design and communications. These help USVs and UUVs to not only perform their designated function, but also to avoid surface and underwater obstacles. The USV/ UUV market is estimated to be $0.77 billion in 2023 and is projected to reach $1.2 billion by 2027, at a compound annual growth rate (CAGR) of 11.1 per cent from 2022 to 2027.



INDIAN NAVAL PLANNERS RECOGNISE THE ADVANTAGES OF UNMANNED CRAFT: NO RISK TO PERSONNEL, GREATER RANGE, SPEED AND ACCURACY OF DATA

Aware of the quickening turn of events, the Indian Navy is aggressively working on its 'unmanned roadmap'. On its wish list are remotely operated and autonomous USVs, UUVs and NUAS. Indian naval planners recognise the advantages of these unmanned craft: reduced manpower requirement and zero risk to personnel, lower operating costs, greater range, improved speed and accuracy of data processing and thus a faster decision cycle. As of now, the objective is to utilise these platforms for ISR purposes, but the option of military use is open too. As with other defence hardware, support and encouragement is being given to domestic firms. To meet its immediate needs, though, the navy is looking at foreign makers. The rising maritime threat from China lends urgency to the matter.